Bell Beam Physics Manual

canonical variables, Lie maps, RF, acceleration

October 30, 2026

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1 Bell architecture

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Bell is a Julia program for symplectic modelling of a magnetic ring. The object of the ring is a product of Lie transformations. Linear optics and higher-order numbers are readings of that product. A truncated Taylor map is an independent check of the same Hamiltonians; it does not define the ring.

This page describes the program: blocks, how they call each other, the two six-dimensional propagators, and the structure file. Physics is in the Beam Physics Manual.

1.1 Blocks

```
flowchart TD
  structureFile[StructureFile] --> elements[Elements]
  elements --> hamiltonians[Hamiltonians]
  hamiltonians --> lie[LieAlgebra]
  hamiltonians --> da[DifferentialAlgebra]
  lie --> factor[OneTurnFactor]
  da --> taylorCheck[TaylorMap_check]
  factor --> linear[Linear]
  factor --> higher[HigherOrder]
  factor --> normalForm[NormalForm]
  factor --> lieTrack[LieTracking]
  hamiltonians --> exact[ExactIntegrator]
  higher --> longitudinal[LongitudinalReduced]
  exact --> acceleration[Acceleration]
  linear --> optimizers[Optimizers]
  higher --> optimizers
  optimizers --> structureFile
  linear --> plotting[Plotting]
  higher --> plotting
  normalForm --> plotting
  lieTrack --> plotting
  exact --> plotting
  longitudinal --> plotting
```

File	Block	Role
src/coordinates.jl	Coordinates	Particle, reference, canonical (x, px, y, py, l,) and the exact map to laboratory angles
src/elements.jl	Elements	Data types only: drift, sector bend, quadrupole, sextupole, thin RF, finite cavity, marker
src/structure.jl	Structure	Named parameters, element cards, sequence of names, Lattice
src/hamiltonians.jl	Hamiltonians	Exact s-Hamiltonian of each element, expanded as a polynomial
src/lie_algebra.jl	Lie algebra	Poisson bracket, Lie series, Dragt–Finn factor, product of factors
src/differential_algebra.jl	Differential algebra	Independent RK4 Taylor map of the same H. Check only
src/linear.jl	Linear	Degree-1 reading: R = exp(:f :), Twiss, dispersion, _c
src/higher_order.jl	Higher order	Readings of the one-turn polynomial: D , , chromaticity
src/normal_form.jl	Normal form	Homological equation on the generators; torus average of
src/tracking.jl	Lie tracking	Flow of each truncated factor; thin RF kick
src/exact_integrator.jl	Exact integrator	Unexpanded H, Yoshida midpoint, finite cavity
src/longitudinal.jl	Longitudinal	Averaged turn-by-turn motion in (,) from already-read
src/acceleration.jl	Acceleration	Reference ramp through the exact integrator
src/optimizers.jl	Optimizers	Optional change of named strengths; never part of the model
src/plotting.jl	Plotting	Figures. No physics
src/symplectic.jl	Symplectic	J S J - S
src/optim_converter.jl	Converter	Writes a structure file from an OptiM .opt

A script loads a structure file and calls readers. An application of a concrete ring lives outside this package and depends on it.

1.2 Where the orders come from

Linear optics and higher-order optics are homogeneous pieces of one one-turn Lie factor

```
F = R   exp(:f :)   exp(:f :)
```

observables(lat) builds that factor once. Twiss and come from degree 1 (F.R). D and come from degree 2 of the polynomial of F. Chromaticity comes from degree 3. The RK4 Taylor map is not used for these numbers.

An element in the structure file is a card of parameters. It is not a procedure that writes into a global map. hamiltonians.jl writes H. The Lie algebra builds a factor from H. Differential algebra repeats the flow of the same H as a truncated polynomial and stops there.

1.3 Two six-dimensional propagators

Both live in (x, px, y, py, l,). Neither lives in the reduced longitudinal model.

Lie tracking applies an already-built Dragt–Finn factor: the flow of `f` , then `f` , then the matrix `R`. A thin RF cavity is a separate canonical kick. There are no terms above the stored polynomial degree. The factor does not depend on `s`, so a finite cavity whose field varies along `s` is not this propagator.

Exact integrator integrates Hamilton's equations with the unexpanded `H` (square roots, not a polynomial) by a symplectic implicit midpoint composed as Yoshida's fourth-order jump. It does not cut a polynomial order; it cuts the step in `s`. It is the propagator for a finite cavity and for a ramp of the reference energy. After the cavity, `renormalize` makes the synchronous particle the new reference.

Full six-dimensional motion is already here. Truncation exists only in Lie tracking and equals the factor order.

The exact integrator reads `H`, not the factor. Reading the factor would turn the finite cavity and the ramp back into a truncated map.

1.4 Reduced longitudinal model

`longitudinal.jl` does not track the ring in 6D. It is the smooth turn-by-turn Hamiltonian in `( , )`:

```
d /dn     = -2  h (    +    ²)
d /dn     = (Z V T / ( ² E)) (sin( _s +  ) - sin  _s)
```

is taken from the higher-order reading of the Lie factor. The ring product does not use this model. Synchrotron tune, separatrix and bucket height are readings of this `H`.

1.5 Structure file

Three parts in one file. The sequence holds names only. Numbers of an element sit on its card. Named parameters sit above the cards so an optimizer changes a name, not a line of the sequence. A quadrupole is written once. The sequence repeats its name as many times as the magnet stands in the ring.

```
reference
  particle deuteron
  kinetic_MeV 270
  harmonic 1

parameters
  L_d = 2.0
  k_f = 0.3

elements
  D1   drift        L = L_d
  QF   quadrupole   L = 0.4    k1 = k_f
  QD   quadrupole   L = 0.4    k1 = -k_f
  B1   sbend        L = 1.5    angle = 0.3926990817

sequence cell
  D1 QF D1 B1 D1 QD D1 B1

sequence
  16 * cell
```

Units are those that already enter the Hamiltonian:

- length: m - bend angle: rad (`deg` on a card is converted when the file is read) - `k1`: $1/m^2$, `k2`: $1/m^3$ - RF voltage: MV per unit charge; phase: rad; harmonic: integer - kinetic energy and rest energy: MeV

Magnetic rigidity is not stored. It is computed from the reference particle. `16 * cell` repeats an already expanded list of names. There are no loops, conditions or matching commands.

`load_structure` builds a `Lattice`. `convert_optim` writes this file from an OptiM `.opt`. The OptiM reader does not assemble a `Lattice`.

1.6 Coordinates kept by Bell

Independent variable: arc length `s` along the reference orbit.

Canonical variables, used in every Hamiltonian and every map:

```
z = (x, px, y, py, l, )
px = Px / p0,   py = Py / p0
l  = -v( ) dt      [m]
   = (p - p0) / p0
```

Laboratory angles `(x', )` are input and output only:

```
x'  = px / ps
    = -h   dt     (late particle: dt > 0    < 0)
```

1.7 Install

From the `Julia` folder:

```
using Pkg
Pkg.activate(".")
Pkg.instantiate()
using Bell
```

2 Beam Physics Manual

Author: Kolokolchikov Sergey Dmitrievich (`sergey.bell13@gmail.com`)

Program blocks and the structure file: Architecture.

Symplectic modelling of a magnetic storage ring with an RF cavity. The independent variable is the arc length s along the reference orbit. All dynamics are written in **canonical** phase-space variables; laboratory angles and RF phase appear only as derived output.

The object of the ring is a product of Lie transformations, not a stack of transport matrices and not a truncated Taylor map. Each element has one Hamiltonian. That H becomes a Dragt–Finn factor $R \circ e^{:f_3:} \circ e^{:f_4:}$. Concatenation in the algebra produces the one-turn factor $\mathcal{F}$. Twiss, slip, chromaticity and the normal form are readings of $\mathcal{F}$. Tracking applies $\mathcal{F}$ to a point. An RK4 polynomial map is kept as a check; it does not define the ring.

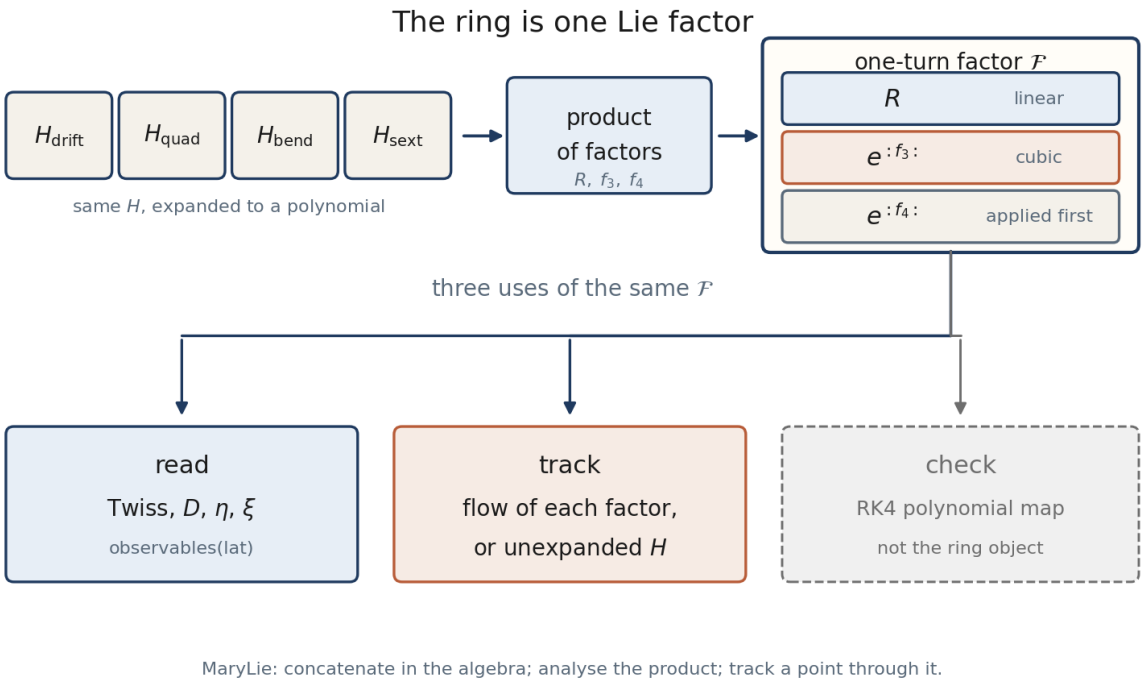


Figure 1: The ring is one Lie factor: Hamiltonians are concatenated in the algebra; the product is read, tracked, or checked.

Linear optics, second-order slip and chromaticity are homogeneous pieces of the same polynomial, not layers derived from one another.

Contents:

1. Canonical phase space
2. Element Hamiltonians
3. Lie product of the ring
4. Linear optics
5. Second-order dispersion and momentum compaction
6. Chromaticity

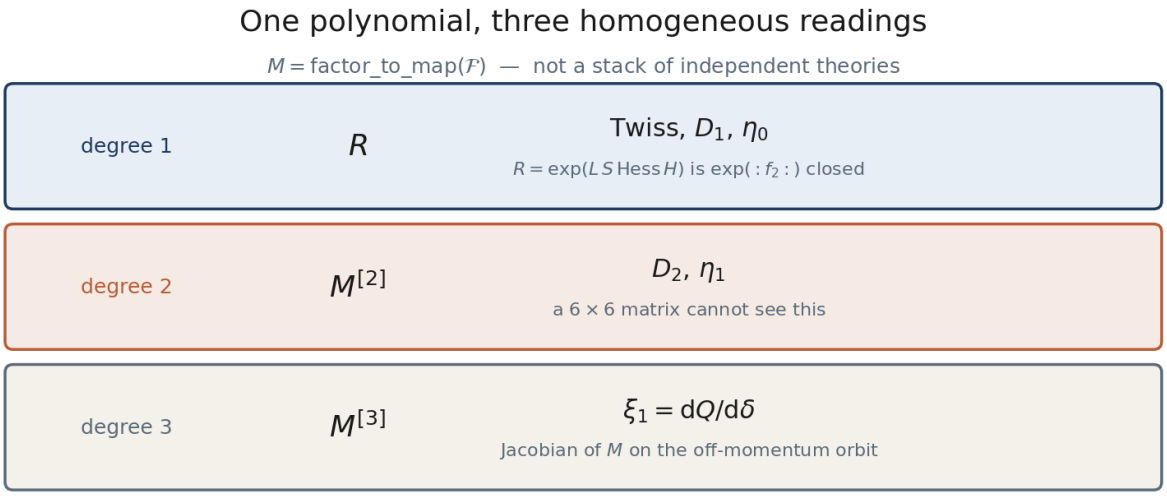


Figure 2: One polynomial, three homogeneous readings: degree 1 gives Twiss and η_0; degree 2 gives D_2 and η_1; degree 3 gives ξ_1.

7. Finite-length RF cavity
8. Acceleration and adiabatic damping
9. Longitudinal motion from the ring average
10. Sextupole families
11. Equilibrium momentum shift
12. Assumptions

Figures use the Nuclotron-based lattice at deuteron kinetic energy 270 MeV unless a ramp from 10 MeV is shown. A separate *Programmer Manual* documents the implementation.

3 Canonical phase space

3.1 Variables

The dynamical state at arc length s is the six-vector

$$z = (x,\, p_x,\, y,\, p_y,\, l,\, \delta).$$

- x, y: transverse displacements from the reference orbit [m]. - $p_x = P_x/p_0$, $p_y = P_y/p_0$: canonical momenta normalised to the reference momentum p_0. - $\delta = (p - p_0)/p_0$: relative *momentum* deviation. - $l = -v(\delta)\,\Delta t$: longitudinal coordinate conjugate to δ, with $\Delta t = t - t_0(s)$ the time lag relative to the reference particle and $v(\delta)$ the particle speed,

$$\frac{v(\delta)}{v_0} = \frac{1+\delta}{\beta_0\sqrt{(1+\delta)^2 + 1/(\beta_0\gamma_0)^2}}.$$

The reference particle has charge Ze, rest energy mc^2, Lorentz factor γ_0, velocity $v_0 = \beta_0 c$ and momentum p_0.

3.2 From $(t, -E)$ to (l, δ)

The s-Hamiltonian is first written with the longitudinal pair $(t, -E)$. A type-2 generating function at fixed s,

$$F_2(t, \delta; s) = -(t - s/v_0)\, E(\delta)/p_0,$$

produces the new coordinate $l = \partial F_2/\partial\delta = -(t - s/v_0)\, v(\delta)$ and adds $\partial F_2/\partial s = E(\delta)/(p_0 v_0)$ to the Hamiltonian. After dropping a constant, that addition is the energy term $e(\delta)$ below. Because l is conjugate to δ, the momentum deviation $\delta = (p - p_0)/p_0$ is the true relative momentum, not an energy offset.

The symplectic form on this space is the standard one,

$$\omega = \mathrm{d}x \wedge \mathrm{d}p_x + \mathrm{d}y \wedge \mathrm{d}p_y + \mathrm{d}l \wedge \mathrm{d}\delta,$$

or in matrix form $S = \mathrm{diag}(S_2, S_2, S_2)$ with $S_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. A map $z \mapsto \mathcal{M}(z)$ is symplectic iff its Jacobian satisfies $J^{\mathsf{T}} S J = S$.

3.3 Output (non-canonical) coordinates

For plots and comparison with laboratory data one uses

$$u = (x,\, x',\, y,\, y',\, \phi,\, \delta), \qquad x' = \frac{p_x}{p_s}, \quad p_s = \sqrt{(1+\delta)^2 - p_x^2 - p_y^2}, \quad \phi = -h\omega_0\Delta t = \frac{2\pi h}{C}\,\frac{l}{v(\delta)/v_0}.$$

$\phi > 0$ means early arrival. The set u is **not** canonical: x' depends on δ and p_y, and ϕ is scaled by $1/v(\delta)$. Symplecticity is therefore checked only in z. The conversion $z \leftrightarrow u$ is exact (nonlinear); truncated polynomial maps apply it as a full Taylor polynomial to the map order.

3.4 Energy term

Every magnetic Hamiltonian uses

$$e(\delta) = \frac{1}{\beta_0}\sqrt{(1+\delta)^2 + \frac{1}{\beta_0^2\gamma_0^2}} - \frac{1}{\beta_0^2},$$

chosen so that the irrelevant constant vanishes at $\delta = 0$ after removal of the constant term of H.

4 Element Hamiltonians

With $p_s = \sqrt{(1+\delta)^2 - p_x^2 - p_y^2}$, curvature $h = 1/\rho$ and normalised multipoles $k_1 = B'/(B\rho)$, $k_2 = B''/(B\rho)$, the s-Hamiltonians (hard-edge, no fringe fields) are

element	$H(z)$
Drift	$e(\delta) - p_s$
Quadrupole	$e(\delta) - p_s + \frac{k_1}{2}(x^2 - y^2)$
Sextupole	$e(\delta) - p_s + \frac{k_2}{6}(x^3 - 3xy^2)$
Sector bend	$e(\delta) - (1+hx)p_s + hx + \frac{1}{2}h^2x^2 + \frac{k_1}{2}(x^2 - y^2)$
Marker	0
Finite RF cavity	$e(\delta) - p_s + A\cos\Phi(l, \delta, s)$ (see Finite-length RF cavity)

For the bend, $\partial H/\partial x|_{z=0} = 0$: the reference arc is a solution of Hamilton's equations.

The bend potential is cut at $hx + \frac{1}{2}h^2x^2$. There are no fringe fields. A gradient in a bend uses the straight-magnet term; curvature corrections of order hk_1x^3 are omitted (the lattice used here has $k_1 = 0$ in every bend).

The Hamiltonian of each magnet is replaced by its Taylor polynomial of total degree 4 (square root and energy expanded). That polynomial is the generator. It is not yet a map.

4.1 Maps from a Hamiltonian

An s-independent Hamiltonian over length L generates the Lie transformation $\exp(-L : H :)$, with

$$: f : g = [f, g] = \sum_i \Big(\partial_{q_i} f \, \partial_{p_i} g - \partial_{p_i} f \, \partial_{q_i} g \Big).$$

How the ring is assembled from these transformations is in Lie product of the ring.

5 Lie product of the ring

The object of the ring is a product of Lie transformations. A generator is a polynomial in the canonical coordinates. The Poisson bracket acts on its coefficients. Substituting truncated maps into one another is a check, not the definition of the one-turn map. The figures in the home page show the product, its three uses, and the homogeneous readings of its polynomial.

5.1 From a Hamiltonian to a factor

Hamilton's equations are $\mathrm{d}z/\mathrm{d}s = [z, H] = -:H:z$. Over a length L with H independent of s the map is the Lie series $\exp(-L:H:)$. The series is cut at total degree 3 in z. The result is written as a Dragt–Finn factor

$$\mathcal{M} = R \circ e^{:f_3:} \circ e^{:f_4:}.$$

Applied to a point, this means: first the time-1 flow of f_4, then of f_3, then the linear matrix R. Each nonlinear flow is a canonical transformation for any generator. Tracking integrates those flows by the implicit midpoint rule, so each factor is symplectic to round-off at any amplitude.

The polynomial expansion of the same factor is symplectic only up to the dropped degree.

5.2 Product of two factors

Two factors are multiplied in the algebra, not by substituting maps. If $\mathcal{M}_a$ is applied first and $\mathcal{M}_b$ second,

$$R = R_b R_a, \qquad f_3 = f_{3a} + f_{3b} \circ R_a, \qquad f_4 = f_{4a} + f_{4b} \circ R_a - \tfrac{1}{2}[f_{3b} \circ R_a,\, f_{3a}].$$

The Poisson bracket of two cubics is quartic. Nested brackets are degree 5 and are dropped. The minus in front of the bracket is the BCH term for the coordinate maps $z \mapsto \exp(:f:)z$; it is fixed by comparing the product of two elements with the Lie series of that pair.

The one-turn magnetic factor is this product over the lattice.

5.3 Thin RF kick

A thin cavity at $\varphi_s = \pi$ is a canonical transformation that fixes the origin: energy changes at fixed arrival time, then δ and l are recomputed. It enters the one-turn product as one more Dragt–Finn factor. The sine of the kick is expanded to the same degree as the magnetic generators. The exact sine is kept only in the independent integrator of the unexpanded Hamiltonian.

A cavity with a net energy gain per turn shifts the origin. The product above is written for maps that fix the origin, so a ramp or a finite cavity with a closed-orbit offset is a separate problem.

5.4 Normal form

The linear part R of the one-turn factor is put into three rotations by a symplectic matrix A, $A^{\mathsf{T}} S A = S$. The generators in the normal coordinates $w = A^{-1}z$ are the pull-backs $f(Aw)$.

A rotation acts on a complex monomial $h_{\mathbf{jn}} = \prod_k (q_k + ip_k)^{j_k}(q_k - ip_k)^{n_k}$ by the phase $e^{-i(\mathbf{j}-\mathbf{n})\cdot\boldsymbol{\mu}}$. The homological operator on a generator is therefore the divisor

$$F - F \circ r(\mu), \qquad F_{\mathbf{jn}} = \frac{-f_{\mathbf{jn}}}{1 - e^{-i(\mathbf{j}-\mathbf{n})\cdot\boldsymbol{\mu}}}.$$

At degree 3, $\mathbf{j} = \mathbf{n}$ is impossible, so in the absence of a third-order resonance the whole cubic generator is removed. At degree 4 the kernel is the span of products of actions; those terms stay as the normal-form Hamiltonian $h_4 = \sum_{i \le j} c_{ij} J_i J_j$. The cubic conjugation contributes $\frac{1}{2}[f_3, F_3]$ to the quartic generator before this step.

The homological operator is applied to the generators. Conjugation of a full polynomial map is not used to build the normal form.

5.5 The torus

After the normal form there are three amplitudes J_x, J_y, J_s and three phases $\theta_x, \theta_y, \theta_s$. The torus of those actions is the set of all triples of phases at those fixed amplitudes: three circles, not a trajectory and not a list of turns.

One turn adds constant angles to the phases and does not change the amplitudes. Laboratory δ is a function of the three phases. The equilibrium momentum is the average of that function over the whole torus at once. Pieces odd in the phases average to zero. What remains is linear in the actions,

$$\delta_{\mathrm{eq}} = \delta^* + d_x J_x + d_y J_y + d_s J_s.$$

For thin RF on the stationary separatrix, δ^* is a numerical zero. The cubic-RF correction in $\hat{\varphi}^2$ is degree J^2 and is not kept.

A long tracking average equals this torus average when the tunes are incommensurate. Tracking does not build the torus.

5.6 Independent checks

Two constructions are not the one-turn Lie product.

1. Hamilton's equations integrated in the truncated polynomial algebra

(a differential-algebra map) and composed by substitution. The coefficient-wise difference with the expanded Lie product is a truncation and step-size error. It is printed; it is not substituted into δ_{eq}.

1. The unexpanded Hamiltonian (the square root is not a polynomial)

integrated by a symplectic step, with an exact thin energy kick at the end of the turn. This is the amplitude-exact tracking standard.

6 Linear optics

The linear map of an element is the closed form of the quadratic Lie transformation $\exp(: f_2 :)$. With $\mathcal{H}_{ij} = \partial^2 H / \partial z_i \partial z_j |_0$,

$$R = \exp(L\, S\, \mathcal{H}).$$

This is not a textbook transport matrix: the Hessian comes from the same H that generates the nonlinear factor. For plots along s the matrices are multiplied element by element. For one-turn optics of the ring, `observables` reads the linear factor $\mathcal{F}.R$ of the Lie product, not a second, independent matrix product.

With $\cos\mu = \frac{1}{2}\mathrm{Tr} M_x$,

$$\beta = \frac{M_{12}}{\sin\mu}, \qquad \alpha = \frac{M_{11} - M_{22}}{2\sin\mu},$$

and the closed dispersion solves $(I - M_4)D = M_{\cdot 6}$. Momentum compaction to first order:

$$\alpha_0 = \frac{1}{\gamma_0^2} - \frac{(l|\delta)_{\text{closed}}}{C} = \frac{1}{C}\oint h\, D\, \mathrm{d}s.$$

Twiss and dispersion propagated in slices close after one turn to machine precision.

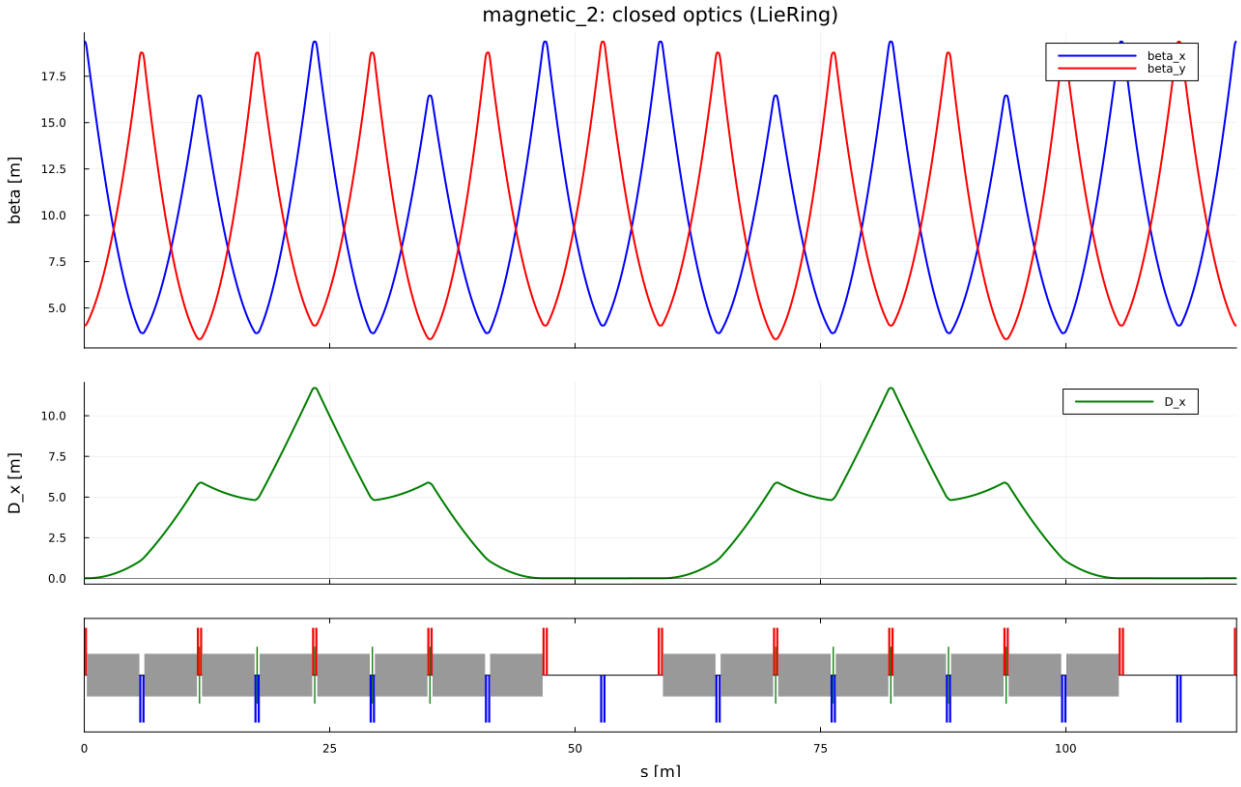


Figure 3: Closed beta functions, horizontal dispersion and magnet layout.

7 Second-order dispersion and momentum compaction

The quadratic piece of the one-turn polynomial. A linear matrix cannot see D_2 or η_1. Both are read from the same Lie product as Twiss; they are not a second optical theory.

7.1 Closed orbit in δ

$$z_\perp(\delta) = D_1\delta + D_2\delta^2 + O(\delta^3),$$

from $M_\perp(z_\perp(\delta), \delta) = z_\perp(\delta)$ order by order.

7.2 Path length and α_i

On the closed orbit the geometric circumference admits the expansion

$$\frac{C(\delta)}{C_0} = 1 + \alpha_0\delta + \alpha_1\delta^2 + \alpha_2\delta^3 + \cdots .$$

With the canonical coordinate l related to time of flight by $l = -v\Delta t$, one turn gives $\Delta l = -C(\delta) + C_0\, v(\delta)/v_0$, hence

$$\alpha_0 = -\frac{l_1}{C} + \frac{1}{\gamma_0^2}, \qquad \alpha_1 = -\frac{l_2}{C} + c_2, \qquad \alpha_2 = -\frac{l_3}{C} + c_3,$$

where $v/v_0 - 1 = c_1\delta + c_2\delta^2 + c_3\delta^3$ and $c_1 = 1/\gamma_0^2$. Equivalently, α_i may be fitted from $C(\delta)/C_0$ evaluated on closed orbits at several δ.

7.3 Slip factor

The relative change of revolution period is the time of flight of one turn. With $l = -v\Delta t$ one has $\Delta T/T_0 = -(l/C)/(v/v_0)$, so on the closed orbit

$$\eta_0 = -\frac{l_1}{C} = \alpha_0 - \frac{1}{\gamma_0^2}, \qquad \eta_1 = -\frac{l_2}{C} + \frac{l_1}{C}\,\frac{1}{\gamma_0^2} = \alpha_1 - \frac{\alpha_0}{\gamma_0^2} + \frac{3\beta_0^2}{2\gamma_0^2} + \frac{1}{\gamma_0^4}.$$

These are the coefficients used in the averaged longitudinal Hamiltonian and in the formula for δ_{eq}. They come from the one-turn Lie product, expanded to a polynomial only to read l_1 and l_2.

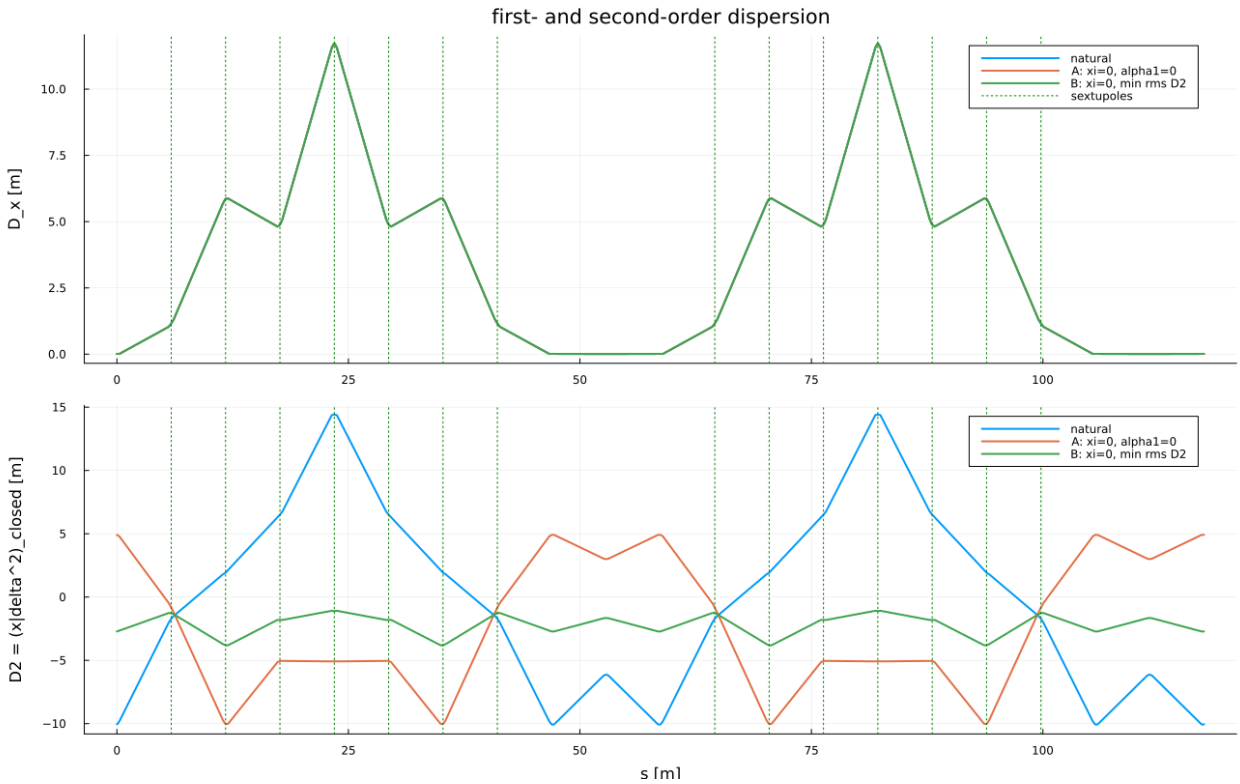


Figure 4: Second-order dispersion along the ring.

8 Chromaticity

Chromaticity is a reading of the one-turn polynomial, not a descendant of the Twiss functions. At fixed δ the transverse motion about $z_\perp(\delta)$ is linearised. The tunes $Q_{x,y}(\delta)$ from $\cos 2\pi Q = \frac{1}{2}\mathrm{Tr}J(\delta)$ expand as

$$Q(\delta) = Q_0 + \xi_1\delta + \tfrac{1}{2}\xi_2\delta^2.$$

Since δ is conserved without RF, these are invariants of the one-turn map. Cross-check: integrate the exact Hamiltonians at several δ and fit the tune of a small betatron oscillation.

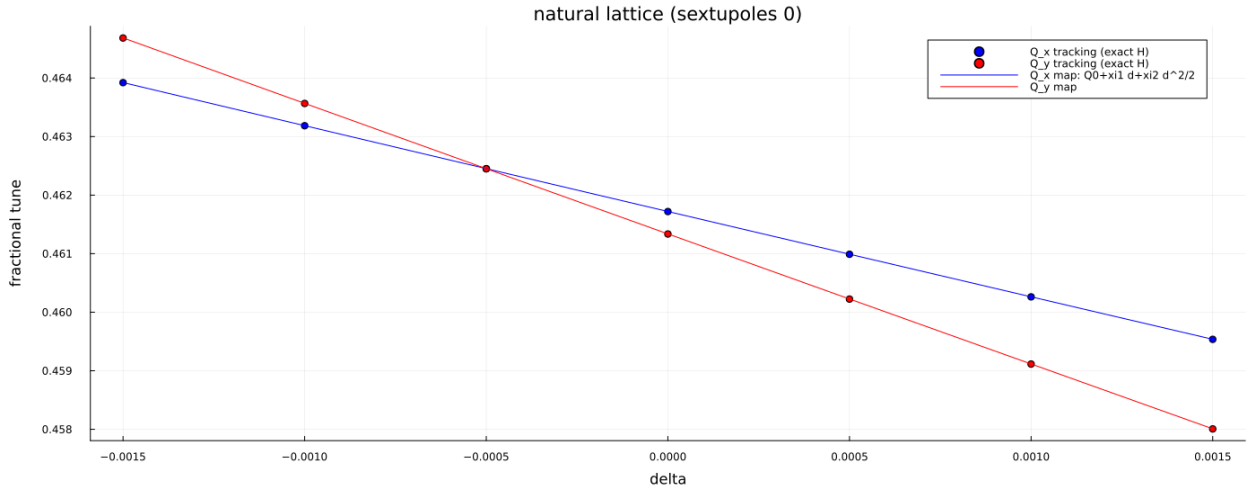


Figure 5: Fractional tunes versus momentum deviation.

9 Fringe fields of dipoles and quadrupoles

The body Hamiltonian of a magnet assumes a field that switches on and off abruptly at the magnet ends. A real field rises over a distance of the order of the gap. The fringe region is described by separate thin maps at the entrance and the exit of the magnet; the body map is unchanged. The lattice keeps the physical description of the magnet: length, bending angle, pole-face angles e_1, e_2, full gap g and fringe integral $K = \mathrm{FINT}$. The edge model is chosen when the ring map is formed.

9.1 Hard edge, linear order

For a pole face rotated by e the field boundary is not perpendicular to the reference orbit. The extra (or missing) triangle of field acts as a thin lens

$$\Delta p_x = h \tan e \; x, \qquad \Delta p_y = -h \tan \psi \; y, \qquad \psi = e - hgK \, \frac{1 + \sin^2 e}{\cos e},$$

with $h = 1/\rho$. The correction $\psi - e$ comes from the finite extent of the fringe: inside it the orbit already bends, so the angle between orbit and field boundary changes. For a sector bend with $e = 0$ and $K = 0$ the linear edge map is the identity.

9.2 Hard edge, nonlinear

The vertical kick depends on the angle between the particle trajectory and the field boundary, not on the angle of the reference orbit. With the slope $x' = p_x/p_s$, $p_s = \sqrt{(1+\delta)^2 - p_x^2 - p_y^2}$, the effective angle is $e + x'$ at the entrance and $e - x'$ at the exit. The edge is the Lie transformation $\exp(- : K :)$ with the generator

$$K_{\mathrm{in}} = -\frac{h \tan e}{2} \, x^2 + \frac{h}{2} \, y^2 \, \tan\bigl(\psi + \arctan x'\bigr), \qquad K_{\mathrm{out}} = -\frac{h \tan e}{2} \, x^2 + \frac{h}{2} \, y^2 \, \tan\bigl(\psi - \arctan x'\bigr).$$

Because the edge is generated by a single polynomial K, the map is symplectic to every order. For $e = 0$ the leading term is $K = \pm h \, y^2 p_x/(2p_s)$. On the dispersive orbit $p_x = D'\delta$, so the vertical focusing of each edge changes linearly with δ: the edges contribute to the vertical chromaticity even when their linear effect vanishes. The same term follows from the longitudinal field $B_s = y \, \partial_s B_y$ of a step profile.

For quadrupoles the leading hard-edge effect is of fourth degree in the generator,

$$K = \pm \frac{k_1}{12(1+\delta)} \left(x^3 p_x + 3xy^2 p_x - 3x^2 y p_y - y^3 p_y \right),$$

(+ entrance, − exit). It changes the amplitude-dependent tune shift and leaves the tunes and the first-order chromaticity unchanged.

9.3 Soft edge

The field on the reference orbit is $B_y = B_0 F(z)$ with a profile F that rises from 0 outside to 1 inside. The effective field boundary (EFB) at $z = 0$ is fixed by equal area,

$$\int_{-W}^{0} F \, dz = \int_{0}^{W} (1 - F) \, dz,$$

so the integrated field and the bending angle are those of the hard-edge magnet. The fringe integral is $gK = \int F(1-F)\, dz$. Profiles:

- Enge: $F = 1/\left(1 + \exp P(-z/g)\right)$, P a polynomial;
- hyperbolic tangent: $F = \frac{1}{2}\left(1 + \tanh(z/\lambda)\right)$,

$gK = \lambda/2$;

- linear ramp of width w: $gK = w/6$;
- tabulated profile from a field map, interpolated and normalised.

The soft region $[-W, W]$ replaces the last length W of the drift and the first length W of the body of the hard-edge layout. The total length and the bending angle are preserved.

Hamiltonian in the fringe

In the Frenet frame of an orbit that follows the local field, $h(s) = h_0 F(s)$,

$$H = e(\delta) - (1+hx)\sqrt{(1+\delta)^2 - (p_x - a_x)^2 - p_y^2} + hx + \frac{h^2x^2}{2} + \frac{b_1}{2}(x^2 - y^2) + \frac{b_2}{6}(x^3 - 3xy^2), \qquad a_x = -\frac{y^2 h_0 F'(s)}{2(1+hx)}.$$

The vector potential a_x gives the longitudinal field $B_s = y\,b'/(1+hx)$ and the second-order vertical dependence $B_y = b - \frac{y^2}{2}\left[b''/(1+hx)^2 - b'h'x/(1+hx)^3\right]$; together they satisfy $\nabla \times B = 0$ to second order in y.

The field boundary is a straight line perpendicular to the field-free orbit, while the frame follows the curved orbit. A point at transverse position x sees the profile at $z = s + x\,\theta(s)$, where θ is the angle between the local orbit and the field-free orbit, $\theta = h_0 \int_{-W}^{s} F$ at the entrance and $\theta = -h_0 \int_{s}^{W} F$ at the exit. Expansion in x gives the gradient $b_1 = h_0 F' \theta$ and the sextupole-like term $b_2 = h_0 F'' \theta^2$. The integrated gradient is

$$\int b_1\, ds = h_0^2 \int F(1-F)\, dz = h_0^2\, gK,$$

which is the vertical focusing of the ψ correction above; in the horizontal plane it cancels the weak-focusing deficit $h_0^2 \int (F^2 - \Theta)\, dz = -h_0^2 gK$ of the shortened field. A soft edge therefore reproduces the linear hard-edge model with the fringe integral of its profile, and its nonlinear terms tend to the nonlinear hard edge as $g \to 0$.

Integration through the fringe

The Hamiltonian depends on s. Three methods give the map of the region:

- **Splitting**: n slices with F, F', F'' replaced by slice values;

each slice is $\exp(-\Delta s : H_k :)$. Second order with one step per slice, fourth order with the Yoshida composition $w_1 = 1/(2 - 2^{1/3})$, $w_0 = 1 - 2w_1$. Every slice is a Lie transformation, so the map is symplectic.

- **Magnus expansion**: on each sub-interval $\exp(- : \Omega :)$ with

$\Omega = \Delta s\, H(s_m)$ (second order) or, with two Gauss nodes, the fourth-order form below. Symplectic.

- **Taylor integration**: the equations of motion integrated in truncated

power series (differential algebra) with step halving until the map changes by less than a tolerance. Not symplectic by construction; the defect is of the order of the tolerance.

Fourth-order Magnus step (H_1, H_2 at the Gauss nodes $s_m \mp \Delta s\sqrt{3}/6$, H_1 first):

$$\Omega = \frac{\Delta s}{2}(H_1 + H_2) - \frac{\sqrt{3}}{12}\Delta s^2\,[H_1, H_2].$$

For profiles with kinks (linear ramp, tabulated data) the derivatives are taken as averages over the slice or step, which keeps $\int F'$ and $\int F''$ exact.

9.4 Validation

- No edge: the ring map is identical to the plain lattice.
- $g \to 0$: the soft-edge tunes and chromaticities approach the

nonlinear hard edge linearly in g.

- Linear focusing of a soft edge equals that of a hard edge with

$K = \int F(1-F)dz/g$.

- Different methods and profiles with the same fringe integral agree to the

accuracy of the slicing.

- $J^{\mathsf{T}}SJ = S$ for the edge maps to round-off (Lie methods).

9.5 Approximations

- The fringe region is treated as straight in the gradient term

b_1, b_2; corrections of order $h_0^2 g^2$ are not included.

- The displacement between the soft-edge and hard-edge reference orbits

(of order $h_0 g^2$) is absorbed in the frames; only maps about the reference are formed.

- Pole-face rotation in a soft edge is the linear wedge at the EFB; the

dependence of the profile on $x \tan e$ is not expanded further.

- Fields are expanded to second order in y; quadrupole soft edges are

not modelled.

10 Finite-length RF cavity

The cavity of length L carries a purely longitudinal on-axis field. With $qA_s/p_0 = -A\cos(\omega t + \psi)$ and $t = s/v_0 - l/v(\delta)$,

$$H_{\mathrm{cav}} = e(\delta) - p_s + A\cos\Phi, \qquad \Phi = \frac{\omega(s - L/2)}{v_0} - \frac{\omega\, l}{v(\delta)} - \phi_s,$$

$$A = \frac{ZV}{p_0 c}\,\frac{c}{\omega L}, \qquad \omega = h\omega_0 = \frac{2\pi h v_0}{C}.$$

Hamilton's equation $\mathrm{d}\delta/\mathrm{d}s = -\partial H/\partial l$ is equivalent to $\mathrm{d}E/\mathrm{d}s = -(ZV/L)\sin\Phi$. Integrated over the cavity the energy gain is

$$\Delta E = ZVT\sin(\phi_s + \phi), \qquad T = \frac{\sin(\theta/2)}{\theta/2}, \quad \theta = \frac{2\pi h L}{C}.$$

H_{cav} depends on s; it is integrated by a symplectic s-dependent scheme (implicit midpoint, Yoshida 4th order). Transverse RF fields are omitted. The thin-kick limit $L \to 0$ recovers an instantaneous energy change $ZV\sin(\phi_s + \phi)$ at fixed (x, p_x, y, p_y, l).

11 Acceleration and adiabatic damping

After the cavity the synchronous particle has $\delta_s \neq 0$. It is made the new reference:

$$p_0' = p_0(1+\delta_s), \quad 1+\delta' = \frac{1+\delta}{1+\delta_s}, \quad p_x' = \frac{p_x}{1+\delta_s}, \quad p_y' = \frac{p_y}{1+\delta_s}, \quad \Delta t' = \Delta t - \Delta t_s.$$

Positions are unchanged. Magnets are assumed to ramp with p_0 (normalised strengths fixed); the RF frequency follows $h\omega_0(p_0)$.

The physical transverse momentum $P_x = p_0 p_x$ is invariant under a longitudinal field, so $p_x \propto 1/p_0$. The Courant–Snyder action in normalised variables then scales as $J \propto 1/p_0$, and betatron amplitudes damp as

$$\frac{A(n)}{A(0)} = \sqrt{\frac{p_{\mathrm{inj}}}{p_0(n)}}.$$

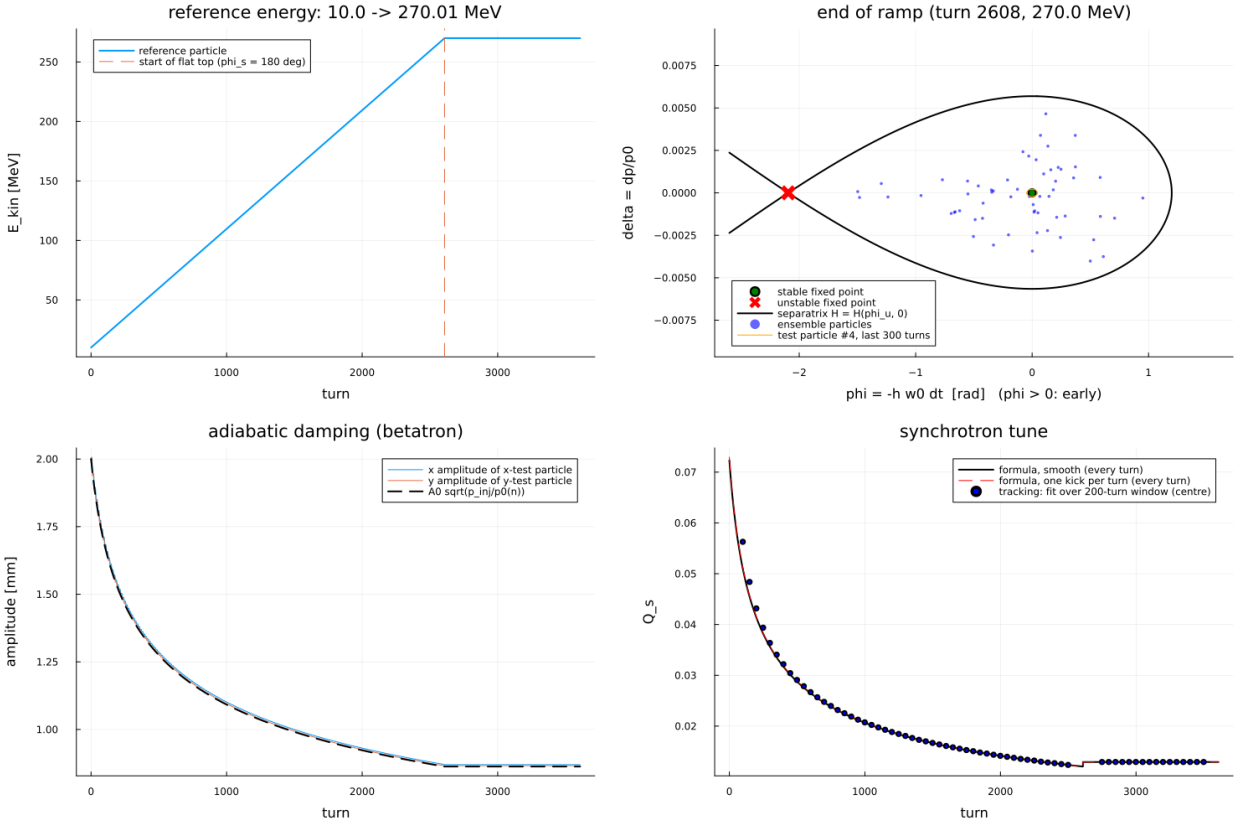


Figure 6: Reference energy, longitudinal phase space with separatrix, adiabatic damping of betatron amplitudes, and synchrotron tune during a ramp.

12 Longitudinal motion from the ring average

12.1 From the 6D Hamiltonian to a 2D synchrotron Hamiltonian

Over one turn the magnetic lattice advances (l, δ) through the element Hamiltonians. The slip coefficients η_0, η_1 are those of the time of flight. A thin RF kick changes δ at fixed arrival time. Coarse-graining to one map per turn in the variables (ϕ, δ) with $\phi = -h\omega_0 \Delta t$ gives

$$\phi_{n+1} = \phi_n - 2\pi h\big(\eta_0 \delta_{n+1} + \eta_1 \delta_{n+1}^2 + \cdots\big),$$
$$\delta_{n+1} = \delta_n + A\big(\sin(\phi_s + \phi_n) - \sin\phi_s\big), \qquad A = \frac{ZVT}{\beta_0^2 E_0}.$$

Treating the turn number as continuous yields the smooth Hamiltonian

$$H(\phi, \delta) = -2\pi h\Big(\frac{\eta_0 \delta^2}{2} + \frac{\eta_1 \delta^3}{3}\Big) + A\big(\cos(\phi_s + \phi) + \phi \sin\phi_s\big),$$

which is the average of the full 6D dynamics over one revolution under the assumptions of a single RF station, no transverse–longitudinal coupling in the averaged equations, and δ small enough that the closed-orbit expansion of the path length converges. Fixed points on $\delta = 0$: stable $\phi = 0$, unstable $\phi_u = \pi - 2\phi_s$. The **separatrix** is the level set $H = H(\phi_u, 0)$; the RF bucket is the region around the origin inside it. Stability requires $\eta_0 \cos\phi_s > 0$.

Small-amplitude synchrotron tune:

$$Q_s = \sqrt{\frac{h\,\eta_0\, ZVT \cos\phi_s}{2\pi\beta_0^2 E_0}} \qquad \text{(smooth)}, \qquad \cos(2\pi Q_s^{\text{kick}}) = 1 - \tfrac{1}{2}(2\pi Q_s)^2 \qquad \text{(one kick/turn)}.$$

12.2 Element-by-element view of one turn

Left: absolute (ϕ, δ) with the analytic separatrix. Right: change relative to the turn start, with fixed axes set by lattice/RF scales —

- horizontal: $\pm 1.5\,|2\pi h\eta_0|\,\delta_{\text{ens}}$ (full-turn phase-slip scale for the largest $|\delta|$ in the ensemble); - vertical: $\pm 1.5\,A$ (RF kick amplitude).

Only the cavity changes δ within a turn; ϕ changes continuously through path length and velocity.

> *Animation (see HTML or GIF file): Ensemble during acceleration (left: absolute; right: δ in units of the current bucket half-height).*

> *Animation (see HTML or GIF file): One turn element by element (left: absolute with separatrix; right: intra-turn change on fixed physical axes).*

13 Sextupole families

A thin sextupole of integrated strength $k_2 L$ contributes to first-order chromaticity

$$\Delta\xi_{x,y} = \pm\frac{1}{4\pi}\sum k_2 L\,\beta_{x,y} D$$

and to second-order dispersion / α_1 through terms $\propto k_2 L D^2$. Three families placed at maxima of D_x, β_x and β_y therefore control $(\xi_x, \xi_y, \alpha_1, D_2)$ nearly independently. Strengths may be chosen to null ξ_x, ξ_y and optionally α_1 or rms D_2; that choice is an optimisation problem, not part of the lattice model. The natural lattice has vanishing sextupole strengths.

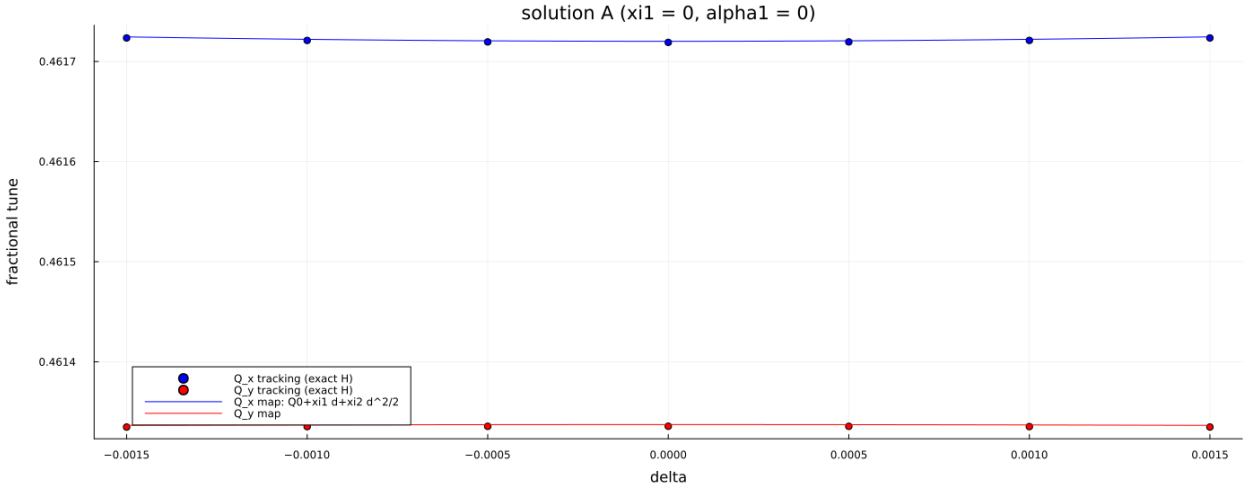


Figure 7: Tunes versus δ after chromaticity correction.

14 Equilibrium momentum shift

A particle that oscillates in a stationary RF bucket ($\varphi_s = \pi$, below transition) does not have mean relative momentum zero. The RF phase returns to itself after a synchrotron period only if the mean time of flight matches the RF period. With a nonlinear slip factor that condition fixes a nonzero mean

$$\delta_{\mathrm{eq}} = -\frac{\eta_1 \langle \delta^2 \rangle + \Lambda}{\eta_0}, \qquad \Lambda = -\frac{2\pi}{C}\big(J_x \xi_x + J_y \xi_y\big).$$

Here η_0, η_1 are the time-of-flight slip coefficients, $\xi_{x,y}$ the linear chromaticities, and $J_{x,y}$ the Courant–Snyder actions. For a linear synchrotron oscillation $\langle \delta^2 \rangle = \delta_m^2/2$. A cubic truncation of the RF waveform multiplies the η_1 piece by $1 - \hat{\varphi}^2/32$ and shifts the synchrotron frequency by $\Omega/\omega_s = 1 - \hat{\varphi}^2/16$. That correction is second order in the actions and is not kept in the normal form below.

The RF is a thin energy kick at one point of the ring. A finite-length cavity has its own closed-orbit offset in δ; that is a separate study.

14.1 From the torus

The one-turn map is the Lie product of the magnetic factors and the thin kick. After the linear symplectic normalisation and the homological removal of non-resonant generators, motion is uniform rotation of three phases at three fixed actions. The torus of those actions is the set of all triples of phases. Laboratory δ is a function on that torus. δ_{eq} is the average of δ over the whole torus:

$$\delta_{\mathrm{eq}} = \delta^* + d_x J_x + d_y J_y + d_s J_s.$$

For thin RF at $\varphi_s = \pi$, $\delta^* = 0$. The coefficients d_k are the quadratic part of the sixth component of the normalising transformation, averaged over angles. They do not use η or ξ. Substituting $\delta_m^2 = 2\sigma_\delta^2 J_s$ into the formula above reproduces them to relative 10^{-6}:

	d_x **[1/m]**	d_y **[1/m]**	d_s **[1/m]**
torus average	0.14665244	0.22328941	2.477064×10^{-3}
formula	0.14665257	0.22328971	2.477064×10^{-3}

Element-by-element symplectic tracking of the unexpanded Hamiltonians, with the exact thin kick, gives the same δ_{eq} to a few parts in 10^3 at $\delta_m \le 10^{-3}$. The residual grows as δ_m^2 (the next order, not kept). A long time average equals the torus average when the tunes are incommensurate. The torus itself is not built from the tracking samples.

The mean energy also receives a kinematic piece from $\langle \delta^2 \rangle$:

$$\Delta E_{\mathrm{eq}} = \beta_s^2 E_s \Big(\delta_{\mathrm{eq}} + \frac{\delta_m^2}{4\gamma_s^2}\Big).$$

15 Assumptions

Remaining modelling assumptions (accepted design choices such as injection energy, RF parameters, CODATA mass, and the interpretation of the lattice file are recorded in the Programmer Manual, not here).

1. Hard-edge magnets, no fringe fields.
2. No space charge, impedances, radiation or alignment errors.
3. RF field purely longitudinal on axis (no transverse RF).
4. Magnets and RF frequency scale with the reference momentum during

acceleration.

1. Single RF station per turn for the averaged longitudinal Hamiltonian.
2. The one-turn object is the Lie product of Dragt–Finn factors, truncated

at degree 4 in the generators. Its polynomial expansion is symplectic only up to the dropped degree. Integration of the unexpanded Hamiltonian is symplectic to round-off.

1. The product is written for maps that fix the origin (thin RF at

$\varphi_s = \pi$). A net energy gain per turn is a separate problem.